BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 13 . MODULE 10 OF 17

Module 10: Endocrine System

Topics: Hormone Mechanisms, Major Endocrine Glands

TOPIC 1 OF 2 . WEEK 13 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Hormone Mechanisms

Steroid vs peptide vs amino-acid-derived hormones, and how each transduces a signal.

The Science

Hormones are signaling molecules released into the blood by endocrine glands. Three chemical classes, two mechanisms.

Lipid-soluble hormones (steroids from cholesterol: cortisol, aldosterone, sex steroids; thyroid hormone): cross the plasma membrane, bind intracellular receptors, act as transcription factors. Effects are slow (hours to days) and long-lasting because they change which proteins the cell is making. Water-soluble hormones (peptides like insulin, growth hormone, ADH; catecholamines like epinephrine): cannot cross the membrane, bind cell-surface receptors, trigger intracellular second messenger cascades (cAMP, IP3/DAG, Ca). Effects are fast (seconds to minutes) and transient.

Feedback: almost every endocrine axis uses long-loop negative feedback. The end-product hormone suppresses the hypothalamus and pituitary that produced the upstream releasing and stimulating hormones. Cortisol suppresses CRH and ACTH. Thyroid hormone suppresses TRH and TSH. Disease often manifests when feedback is broken: primary failure (gland itself broken; downstream hormone low, upstream high because no negative feedback), secondary failure (pituitary broken; both upstream and downstream low), tertiary failure (hypothalamus broken). Pattern recognition here is high-yield.

Teaching This Topic

Before the video (drawing prompt)

Have students sketch the hypothalamus-pituitary-target gland axis as a feedback loop. Where does the end-hormone send its negative feedback signal?

Common misconceptions to address

- Students think hormones act on every cell. They act only on cells that have the right receptor.
- Steroid versus peptide mechanism is sometimes blurred. Steroids = intracellular receptor + transcription. Peptides = surface receptor + second messenger.
- Negative feedback in endocrine systems is sometimes thought to be optional. It is the default; positive feedback is the rare exception (LH surge before ovulation).

Order of operations (what to teach first)

Three chemical classes, then two mechanisms, then feedback patterns and the primary/secondary/tertiary failure logic.

Self-test prompts

- Where are steroid hormone receptors located?
- Why must insulin be injected and not swallowed?
- What does it mean when a patient has high ACTH and low cortisol?

Why This Matters to Your Patients

Primary hypothyroidism (Hashimoto): low T4, high TSH. Secondary hypothyroidism (pituitary failure): low T4, low TSH. Same low T4, different cause. Cushing syndrome: high cortisol producing hyperglycemia, central obesity, immunosuppression, hypertension. Addison disease: primary adrenal failure with low cortisol, high ACTH, skin hyperpigmentation. Diabetes insipidus: low ADH, dilute high-volume urine. SIADH: high ADH, concentrated low-volume urine, dilutional hyponatremia. Every one of these stories is a hormone level plus a feedback consequence.

TOPIC 2 OF 2 . WEEK 13 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Major Endocrine Glands

Pituitary, thyroid, parathyroid, adrenal, pancreas, gonads: who makes what and what it does.

The Science

The pituitary is the master, but the hypothalamus is its boss. The hypothalamus releases small peptides (CRH, TRH, GnRH, GHRH, etc.) that travel via a tiny portal system to the anterior pituitary, which then releases the trophic hormones: ACTH, TSH, FSH, LH, GH, prolactin. The posterior pituitary stores and releases ADH and oxytocin made in the hypothalamus.

Thyroid: T3 and T4 set metabolic rate; calcitonin lowers blood calcium (modest in adults). Parathyroids (four small glands posterior to the thyroid) release PTH, which raises blood calcium by osteoclast activation, renal calcium retention, and vitamin D activation. Adrenal cortex (three zones, top to bottom): zona glomerulosa makes aldosterone (Na retention), zona fasciculata makes cortisol (stress and metabolism), zona reticularis makes androgens. Adrenal medulla makes epinephrine and norepinephrine (sympathetic effector).

Pancreas islets: alpha cells make glucagon (raises blood glucose), beta cells make insulin (lowers blood glucose), delta cells make somatostatin (regulator). Gonads: testes make testosterone; ovaries make estrogen and progesterone. Each gland is a story; pair each hormone with its main job in one sentence and students can build a matrix they will use for the rest of healthcare.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a body outline and place every major endocrine gland with one hormone and one job per gland. They have to commit before the video.

Common misconceptions to address

- Students confuse hypothalamus and pituitary. Hypothalamus is the upstream controller; pituitary is the executive that fires hormones into the blood.
- ADH and oxytocin are sometimes confused. ADH = water retention; oxytocin = labor, milk letdown, social bonding.
- The adrenal cortex and medulla are different tissues with different embryonic origins, despite sharing one capsule.

Order of operations (what to teach first)

Hypothalamus-pituitary axis, then thyroid/parathyroid, then adrenals, then pancreas, then gonads.

Self-test prompts

- Name the gland that makes ADH and where it is released.
- What three things does PTH do to raise blood calcium?
- Which adrenal zone makes aldosterone and what does aldosterone do?

Why This Matters to Your Patients

Type 1 diabetes: autoimmune destruction of pancreatic beta cells; no insulin; hyperglycemia, ketoacidosis without exogenous insulin. Type 2 diabetes: insulin resistance plus eventual beta cell failure. Hyperthyroidism (Graves): heat intolerance, weight loss, tremor, palpitations. Hypothyroidism: cold intolerance, weight gain, fatigue, bradycardia. Pheochromocytoma: a catecholamine-secreting adrenal tumor producing episodic hypertension, headache, sweating, palpitations. Hyperparathyroidism: high calcium producing kidney stones, bone loss, neuropsychiatric symptoms. Each gland has its overproduction and underproduction story; nurses need both for any patient with endocrine symptoms.